Convexity Interpolation Notes

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Recalling the Construciton of a Convex Interpolant

If the inequality
$$y_j \geq y_i + g_i^\top (x_j - x_i) \quad \text{for all } i, j$$
is satisfied, then one can explicitly construct a convex function $f(x)$ that interpolates the data.

Each pair (x_i, y_i) along with its subgradient g_i defines an affine function:
$$f_i(x) = y_i + g_i^\top (x - x_i).$$

If the inequality condition holds, we can take the maximum over these affine functions to define a convex interpolant:
$$f(x) = \max_{i=1,\ldots,n} \left\{ y_i + g_i^\top (x - x_i) \right\}.$$

This $f(x)$ is convex because the pointwise maximum of affine functions is convex. Moreover, it satisfies $f(x_i) = y_i$, hence it interpolates the given data.

Convexity via First-Order Inequality: Two Applications

The first-order condition for convexity is given by:
$$f(x_j) \geq f(x_i) + \langle g_i, x_j - x_i \rangle$$

This inequality ensures that the affine approximation $f(x_i) + \langle g_i, x - x_i \rangle$ lies below the function $f(x)$, i.e., that f is convex. It plays a central role in two distinct applications.

Case 1: Constructing a Convex Function from Data

Suppose we are given points (x_i, y_i). To construct a convex function f that interpolates these points, the standard method is:
$$f(x) = \max_{i=1,\ldots,n} \left\{ y_i + g_i^\top (x - x_i) \right\}$$

Each term $y_i + g_i^\top (x - x_i)$ is an affine function. Since the pointwise maximum of affine functions is convex, $f(x)$ will be convex if the subgradients g_i satisfy the subgradient condition:
$$y_j \geq y_i + g_i^\top (x_j - x_i) \quad \text{for all } i, j$$

Case 2: Verifying Convexity of a Function Sum

Now consider two fixed convex functions f and f'. The goal is to find the smallest value of β such that the function

$$f(x) + \beta f'(x)$$

is still convex. For this, we assume the subgradients g_i come from f, and define:

$$y_i = f(x_i), \quad y_i' = f'(x_i)$$

The convexity condition becomes:

$$f(x_j) + \beta f'(x_j) \geq f(x_i) + \beta f'(x_i) + g_i^\top (x_j - x_i)$$

This is rearranged to isolate β, and a linear program is solved to find the largest such value for each i. We define:

$$\beta_i = \max \left\{ \beta : y_j - y_i \geq g_i^\top (x_j - x_i) + \beta(y_j' - y_i') \right\}$$

Then the overall threshold is given by:

$$\beta^* = \min_i \beta_i.$$

Though the purposes differ, both applications are grounded in the same first-order condition for convexity.

Convexity Interpolation

Tony Deng

The question we try to solve is: given two convex functions $f, f' : \mathbb{R}^d \to \mathbb{R}$, we want to find the minimum β such that $f + \beta f'$ is still convex.
A discrete version of this problem based on point interpolation can be formulated as follows:
Given a list of triples (x_i, y_i, y_i') where $x_i \in \mathbb{R}^d, y_i, y_i' \in \mathbb{R}$, we want to find the minimum β such that $y_i + \beta y_i'$ interpolate a convex function. This problem can be further reformulated as the following linear program

$$\begin{cases} \min \beta \\ y_j - y_i + \beta(y_j' - y_i') \geq (x_j - x_i)^T g_i \end{cases}$$

Why We Check Convexity of $f + \beta f'$

Slide 10 states that we need to impose the convexity of the potential. This is because we want the resulting flow map to be the gradient of a convex function — a key structural requirement in many settings, including optimal transport.

To enforce this, we study whether the function $f + \beta f'$ remains convex for a given scalar β. This expression represents a convex base potential f perturbed in the direction of another convex function f', scaled by β. We are trying to find the smallest such $\beta \geq 0$ that preserves convexity.

In the flow defined by:

$$z^{k+1} = z^k + \beta^k \nabla_x F(z^k) = z^0 + \sum_{i=1}^{k} \nabla_x F(z^i),$$

the total displacement is a sum of gradients. If each $\beta^k > 0$, the potential remains convex since sums of convex functions are convex. However, if any $\beta^k < 0$, we must verify that the resulting potential is still convex.

Therefore, checking the convexity of $f+\beta f'$ ensures that the accumulated flow remains valid — i.e., generated by a convex potential — even when perturbations are introduced.

Convexity via First-Order Interpolation Inequality

The key convexity condition (in subgradient form) as described in the interpolation problem is:

$$\boxed{f(x_j) \geq f(x_i) + \langle g_i,\ x_j - x_i\rangle}$$

where g_i is a subgradient of f at the point x_i. The notation $\langle \cdot,\ \cdot\rangle$ denotes the dot product (inner product) between vectors in $\mathbb{R}^d$.

In our setting, we are considering values of the function $f + \beta f'$ evaluated at points x_i, so we want the following inequality to hold:

$$y_j + \beta y'_j \geq y_i + \beta y'_i + \langle g_i,\ x_j - x_i\rangle$$

Rewriting this by moving terms to one side:

$$y_j - y_i + \beta(y'_j - y'_i) \geq \langle x_j - x_i,\ g_i\rangle$$

This is the constraint used in the linear program.

Final Linear Program

We minimize β, subject to the constraint:

$$y_j - y_i + \beta(y'_j - y'_i) \geq (x_j - x_i)^\top g_i \quad \text{for all } i, j$$

This corresponds to finding the smallest β such that adding $\beta f'$ to f still results in a convex function over the observed sample points.

One idea on OT

We need to impose the convexity of the potential $\to$ we want the map to be the gradient of a convex function. If we build the flow

$$z^{k+1} = z^k + \beta^k \nabla_x F(z^k) = z^0 + \sum_{i=1}^{k} \nabla_x F(z^i)$$

Missing a beta_i here

with $z^0 = x$ the starting position, $\beta \in R$, and F convex, then the convexity of the potential depends on the sign of β.

- When β^i is positive there is no problem (sum of convex functions is still convex)
- If β is negative we need to pay attention
- We only have the value $\sum_{i=1}^{k} \nabla_x F(z^i)$ at the sample points z^i
- A condition that $\sum_{i=1}^{k} \nabla_x F(z^i)$ should satisfy is that, at least, should interpolate a convex function

Baptista, Trigila, Wang PolyMathJr 2025: ML project 10 / 13

Note: If all $\beta^k = 0$, then the flow is frozen — that is, the update rule

$$z^{k+1} = z^k + \beta^k \nabla_x F(z^k)$$

reduces to $z^{k+1} = z^k$ at every step. This means the iterates do not change and the map constructed is simply the identity. The potential function F is still present, but its gradient does not influence the flow.

Since by assumption, this minimum occurs when $\beta < 0$, we can also formulate the problem by replacing β by $-\beta$,

$$\begin{cases} \max \beta \\ y_j - y_i \geq (x_j - x_i)^T g_i + \beta(y'_j - y'_i) \end{cases}$$

Reformulating the Problem When $\beta < 0$

Since, by assumption, the optimal value of β occurs when $\beta < 0$, we can simplify the optimization by substituting $\beta \leftarrow -\beta$. This turns the minimization of a negative quantity into a maximization of a positive one.

Starting from the original constraint:

$$y_j - y_i + \beta(y'_j - y'_i) \geq (x_j - x_i)^T g_i$$

We substitute $\beta \leftarrow -\beta$, which gives:

$$y_j - y_i - \beta(y'_j - y'_i) \geq (x_j - x_i)^T g_i$$

Rewriting this:

$$y_j - y_i \geq (x_j - x_i)^T g_i + \beta(y'_j - y'_i)$$

This leads to the following reformulated linear program:

$$\begin{cases} \max \beta \\ \text{subject to} \quad y_j - y_i \geq (x_j - x_i)^T g_i + \beta(y'_j - y'_i) \quad \text{for all } i, j \end{cases}$$

This is equivalent to the original problem, but posed as a maximization of a positive β, which can simplify interpretation and computation.

By using matrix notation, we construct matrices

$$A_i = \begin{bmatrix} (x_1 - x_i)^T & y'_1 - y'_i \\ \vdots & \\ (x_n - x_i)^T & y'_n - y'_i \end{bmatrix}, b_i = \begin{bmatrix} y_1 - y_i \\ \vdots \\ y_n - y_i \end{bmatrix}$$

Matrix Formulation of the Constraints

To simplify the convexity constraints across all points j, we express them using matrix notation.

Recall the original pointwise convexity constraint:

$$y_j - y_i + \beta(y'_j - y'_i) \geq (x_j - x_i)^T g_i \quad \text{for all } j$$

We rewrite this in matrix form as:

$$A_i \begin{bmatrix} g_i \\ \beta \end{bmatrix} \leq b_i$$

where the matrices are defined as follows:

$$A_i = \begin{bmatrix} (x_1 - x_i)^T & y'_1 - y'_i \\ \vdots & \vdots \\ (x_n - x_i)^T & y'_n - y'_i \end{bmatrix}, \qquad b_i = \begin{bmatrix} y_1 - y_i \\ \vdots \\ y_n - y_i \end{bmatrix}$$

Each row of A_i corresponds to a constraint for a specific index j. The first component $(x_j - x_i)^T$ multiplies the subgradient g_i, and the second component $y'_j - y'_i$ multiplies β.

The vector b_i captures the right-hand sides $y_j - y_i$ of all constraints for fixed i. Together, this matrix formulation compactly encodes all the linear inequalities needed to ensure that the values $y_i + \beta y'_i$ interpolate a convex function.

Let $c = (0, 0, ..., 0, 1)$ be the $d+1$ dimensional vector with 1 in its last coordinate and 0 elsewhere. Then the linear problem becomes

$$\begin{cases} \max c^T x \\ A_i x \geq b_i \end{cases}$$

Since the threshold for β must satisfies the criterion that the linear program must be feasible for all i, we define

$$\beta_i = \begin{cases} \max c^T x \\ A_i x \geq b_i \end{cases}$$

And the threshold is

$$-\beta = -\min_i \beta_i$$

Dimension Check for the Constraint $A_i \begin{bmatrix} g_i \\ \beta \end{bmatrix} \leq b_i$

We verify that the matrix multiplication in the constraint

$$A_i \begin{bmatrix} g_i \\ \beta \end{bmatrix} \leq b_i$$

is dimensionally consistent.

- The vector $g_i \in \mathbb{R}^d$, and $\beta \in \mathbb{R}$. Therefore, the stacked vector is

$$\begin{bmatrix} g_i \\ \beta \end{bmatrix} \in \mathbb{R}^{d+1}.$$

- The matrix A_i is defined as

$$A_i = \begin{bmatrix} (x_1 - x_i)^\top & y'_1 - y'_i \\ \vdots & \vdots \\ (x_n - x_i)^\top & y'_n - y'_i \end{bmatrix} \in \mathbb{R}^{n \times (d+1)}.$$

  Each row consists of a d-dimensional row vector plus one scalar, making $d+1$ columns total, and there are n rows (one for each j).

- The right-hand side vector b_i is

$$b_i = \begin{bmatrix} y_1 - y_i \\ \vdots \\ y_n - y_i \end{bmatrix} \in \mathbb{R}^n.$$

Therefore, the matrix-vector product is

$$A_i \begin{bmatrix} g_i \\ \beta \end{bmatrix} \in \mathbb{R}^n,$$

which is compatible with the inequality $A_i[g_i; \beta] \leq b_i$, since both sides are in $\mathbb{R}^n$.

Step-by-Step Explanation of the Threshold Computation for β

Understanding the Vector c and Its Role in the Linear Program

We are solving a linear program where the decision variable is a vector

$$x = \begin{bmatrix} g_i \\ \beta \end{bmatrix} \in \mathbb{R}^{d+1}.$$

- The first d entries of x represent the subgradient $g_i \in \mathbb{R}^d$.
- The last entry of x is the scalar $\beta \in \mathbb{R}$.

So the full decision variable x bundles both g_i and β into a single $(d+1)$-dimensional vector.

Extracting and Optimizing β

We want to optimize the scalar β, which is the last coordinate of the vector x. To do this, we define a vector

$$c = (0, 0, \dots, 0, 1) \in \mathbb{R}^{d+1},$$

which is a row vector with:

- Zeros in the first d coordinates,
- A 1 in the last coordinate (which corresponds to β).

Why This Works

When we compute the dot product $c^\top x$, we get:

$$c^\top x = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{bmatrix} g_i \\ \beta \end{bmatrix} = \beta.$$

Therefore, maximizing $c^\top x$ is equivalent to maximizing β.

The Linear Program for each i

Each subproblem (for fixed i) becomes:

$$\begin{aligned} \text{maximize} \quad & c^T x \\ \text{subject to} \quad & A_i x \leq b_i \end{aligned}$$

This finds the maximum allowable value of β that satisfies the convexity condition for fixed point x_i.

Call this optimal value β_i.

Combine all constraints

The convexity condition must hold for all i, so we take:

$$\beta^* = \min_i \beta_i$$